THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonal of a rectangle is 13 cm and one side is 5 cm. Its area is: 1

(a) 60 cm^2 **(b)** 65 cm^2 **(c)** 72 cm^2 **(d)** 70 cm^2

2. The diagonals of a rhombus are 14 cm and 48 cm. Its perimeter is: 1

(a) 125 cm **(b)** 150 cm **(c)** 104 cm **(d)** 100 cm

3. The diagonals of a rhombus are 48 cm and 90 cm. Its area is: 1

(a) 2160 cm^2 **(b)** 2170 cm^2 **(c)** 2208 cm^2 **(d)** 2250 cm^2

4. The diagonals of a quadrilateral are 28 cm and 22 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 102 cm **(b)** 50 cm **(c)** 75 cm **(d)** 150 cm

5. In a parallelogram, two opposite angles are $(3x + 20)^\circ$ and $(x + 130)^\circ$. The value of x is: 1

(a) 55 **(b)** 57 **(c)** 54 **(d)** 56

6. In $\triangle ABC$, D and E are the midpoints of AB and AC and $DE = 19$ cm. Then BC is: 1

(a) 42 cm **(b)** 38 cm **(c)** 40 cm **(d)** 57 cm

7. The angles of a quadrilateral are in the ratio $3 : 6 : 5 : 1$. The largest angle is: 1

(a) 154° **(b)** 134° **(c)** 144° **(d)** 164°

8. One angle of a parallelogram exceeds another adjacent angle by 40°. The smaller angle is: 1

(a) 70° **(b)** 65° **(c)** 90° **(d)** 80°

9. The diagonal of a rectangle is 39 cm and one side is 15 cm. Its area is: 1

(a) 550 cm^2 **(b)** 576 cm^2 **(c)** 555 cm^2 **(d)** 540 cm^2

10. The diagonals of a rhombus are 30 cm and 72 cm. Its perimeter is: 1

(a) 156 cm **(b)** 195 cm **(c)** 234 cm **(d)** 160 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A rectangle has a diagonal of 25 cm and one side 7 cm. Find its other side, area and perimeter. 2

12. The diagonals of a rhombus are 28 cm and 96 cm. Find its side, perimeter and area. 2

13. One angle of a parallelogram is 50° more than its adjacent angle. Find all the angles. 2

14. A rectangle has a diagonal of 26 cm and one side 10 cm. Find its other side, area and perimeter. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The diagonal of a square is 38 cm. Find its area and its side in terms of $\sqrt{2}$. 3

16. In rhombus ABCD, $\angle A = 54°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

17. In parallelogram ABCD, $AB = (2x + 3)$ cm, $CD = (x + 9)$ cm and $BC = (x + 5)$ cm. Find x, the sides and the perimeter. 3

OR

The diagonal of a square is 8 cm. Find its area and its side in terms of $\sqrt{2}$. 3

18. In rhombus ABCD, $\angle A = 62°$. Find $\angle B$, and the angles $\angle OAB$ and $\angle OBA$ where O is the point of intersection of the diagonals. 3

OR

In parallelogram ABCD, $AB = (2x + 1)$ cm, $CD = (x + 11)$ cm and $BC = (x + 2)$ cm. Find x , the sides and the perimeter. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. ABCD is a parallelogram with $AB = 13$ cm, $AD = 14$ cm and diagonal AC with $AC^2 = 365$ cm^2. Show that ABCD is a rectangle. 5

OR

The perimeter of $\triangle ABC$ is 116 cm. D, E, F are the midpoints of its sides. Find the perimeter of $\triangle DEF$ and the length of each of its sides in terms of the sides of $\triangle ABC$. 5

20. ABCD is a parallelogram with $AB = 7$ cm, $AD = 13$ cm and diagonal AC with $AC^2 = 218$ cm^2. Show that ABCD is a rectangle. 5

OR

The perimeter of $\triangle ABC$ is 84 cm. D, E, F are the midpoints of its sides. Find the perimeter of $\triangle DEF$ and the length of each of its sides in terms of the sides of $\triangle ABC$. 5

END OF QUESTION PAPER Total: 40 Marks

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